Template Screening Report (v5.7.1 - Multi-Source AI Evaluator)

Journal Template: Jurnal Teknologi

Submission No: 26791

Author Name: Kadum et al.

Article Title: Desulfurization of Heavy Naphtha with synthesized active carbon (PDAC) and impregnated (Cu■Cl): Optimization, Kinetic, and Isotherm Studies

Source File: PAPAR FINAL.docx

Screened PDF: PAPAR FINAL.pdf

Screening Date: 2026-05-21 09:39:08

Detection Source Debug Report

Item	DOC Detection	PDF Detection	AI Detection	Final Source
Article Title	NO	YES	YES	PDF + AI
Authors	NO	YES	YES	PDF + AI
Affiliations	NO	YES	YES	PDF + AI
Corresponding Author	YES	YES	SKIPPED	DOC + PDF
Graphical Abstract	YES	YES	SKIPPED	DOC + PDF
English Abstract	YES	YES	SKIPPED	DOC + PDF
Keywords	YES	YES	SKIPPED	DOC + PDF
Malay Abstract (Abstrak)	NO	YES	YES	PDF + AI
Kata Kunci	NO	NO	NO	None
Introduction	YES	NO	YES	DOC + AI
Methodology	YES	YES	SKIPPED	DOC + PDF
Results and Discussion	NO	NO	NO	None
Conclusion	NO	NO	NO	None
Acknowledgement	NO	NO	SKIPPED	None
References	NO	NO	SKIPPED	None

Summary: DOC detected 6 items | PDF detected 9 items | AI confirmed 5 items

The system uses DOC + PDF + AI voting logic. AI serves as a third checker when DOC and PDF conflict. AI is skipped when DOC and PDF already agree.

Executive Summary

Overall Decision: REJECT

Main Reason: The manuscript is missing required template components or critically mismatched.

Critical Failed Items:

• Results and Discussion

• Conclusion

• References

Items Needing Review:

• Article Title

• Authors

• Affiliations

• Malay Abstract (Abstrak)

• Introduction

Suggested Action: Author should revise the manuscript using the official Jurnal Teknologi template before proceeding.

Overall Result

Metric	Score / Value
Final Status	REJECT
Final Score	73/100
Structure Score	40%
Layout Score	89%
Formatting Score	100%
Detection Method	DOCX-First + Multi-Source AI (v5.7.1)
Decision	REJECT

How Items Are Evaluated

DOC + PDF + AI SKIPPED	PASS	Both deterministic detectors agree
DOC + PDF + AI CONFIRMED	PASS	All three confirm
DOC + AI	NEEDS REVIEW	DOC+AI confirm, but PDF layout conflicts
PDF + AI	NEEDS REVIEW	PDF+AI confirm, but DOC source extraction fails
DOC only	NEEDS REVIEW	DOC detects, but PDF+AI don't confirm
PDF only	NEEDS REVIEW	PDF detects, but DOC+AI don't confirm
AI only	NEEDS REVIEW	Only AI confirms (deterministic detectors failed)
None	FAILED	All three fail to detect required item

Status Explanation

PASS — Item follows the template structure, position, and formatting.

WARNING/NEEDS REVIEW — Item exists but evidence is inconsistent or layout differs from template.

FAILED — Required item truly missing after DOC/PDF/AI checks.

Item Screening Details

Item	DOC	PDF	AI	Final Source	Status	Reason

Article Title	NO	YES	YES	PDF + AI	WARNING	Title detected by PDF and AI, but DOC source text did not co
Authors	NO	YES	YES	PDF + AI	WARNING	Authors detected by PDF and AI, but DOC source text did not
Affiliations	NO	YES	YES	PDF + AI	WARNING	Affiliations detected by PDF and AI, but DOC source text did
Corresponding Author	YES	YES	SKIPPED	DOC + PDF	PASS	Corresponding_Author confirmed by both DOC and PDF detection
Graphical Abstract	YES	YES	SKIPPED	DOC + PDF	PASS	Graphical_Abstract confirmed by both DOC and PDF detection.
English Abstract	YES	YES	SKIPPED	DOC + PDF	PASS	Abstract confirmed by both DOC and PDF detection.
Keywords	YES	YES	SKIPPED	DOC + PDF	PASS	Keywords confirmed by both DOC and PDF detection.
Malay Abstract (Abstrak)	NO	YES	YES	PDF + AI	WARNING	Abstrak detected by PDF and AI, but DOC source text did not
Kata Kunci	NO	NO	NO	None	FAILED	Kata_Kunci not found by DOC, PDF, or AI. [AI: weak_evidence,
Introduction	YES	NO	YES	DOC + AI	WARNING	Introduction confirmed by DOC and AI, but PDF/layout detecti
Methodology	YES	YES	SKIPPED	DOC + PDF	PASS	Methodology confirmed by both DOC and PDF detection.
Results and Discussion	NO	NO	NO	None	FAILED	Results_Discussion not found by DOC, PDF, or AI. [AI: weak_e
Conclusion	NO	NO	NO	None	FAILED	Conclusion not found by DOC, PDF, or AI. [AI: weak_evidence,
Acknowledgement	NO	NO	SKIPPED	None	PASS	Acknowledgement not found by DOC, PDF, or AI.
References	NO	NO	SKIPPED	None	FAILED	References not found by DOC, PDF, or AI.

Detailed Findings

Article Title [title]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Article Title' section must exist.
Found: Matched heading: "DESULFURIZATION OF HEAVY NAPHTHA WITH" | Detection method: score_ranking
Confidence: 0.85
Detection Method: score_ranking
DOC Detection: NO
PDF Detection: YES
AI Detection: YES
AI Confidence: 0.85
AI Evidence: PDF snippet shows a complete and formatted title section with all expected elements. DOC snippet does not contain the title section, only partial meta
Evidence Sources: PDF + AI
Final Source Used: PDF + AI
Rejected false positives: Jurnal Teknologi, Full Paper, Article history
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, top area
Found: Page 1, left column, upper area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.10).

C. Formatting Evaluation

Expected: Heading should be in uppercase style.
Expected: Heading should use bold font.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Authors [authors]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Authors' section must exist.
Found: Matched heading: "DESULFURIZATION OF HEAVY NAPHTHA WITH" | Detection method: score_ranking
Confidence: 0.55
Detection Method: score_ranking
DOC Detection: NO
PDF Detection: YES
AI Detection: YES
AI Confidence: 0.75
AI Evidence: PDF snippet shows authors section with names (Fatima K.Hamza, Abbas K.Mohammad), affiliations (College of Engineering, Department of Chemistry, Univer
Evidence Sources: PDF + AI
Final Source Used: PDF + AI
Rejected false positives: Article history, Received, Accepted
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, left column, upper area
Position Check: PASS
Reason: Position matches expected template region.

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Affiliations [affiliations]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Affiliations' section must exist.
Found: Matched heading: "SYNTHESIZED ACTIVE CARBON (PDAC) AND IMPREGNATED (CU■CL): OPTIMIZATION, KINETIC, AND ISOTHERM STUDI" | Detection method: score_rank
Confidence: 0.70
Detection Method: score_ranking
DOC Detection: NO
PDF Detection: YES
AI Detection: YES
AI Confidence: 0.85
AI Evidence: PDF snippet includes author names, locations, and institutional affiliations (College of Engineering, Department of Chemistry, University of Al-Qadisi
Evidence Sources: PDF + AI
Final Source Used: PDF + AI
Rejected false positives: Accepted
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, left column, upper area
Position Check: PASS
Reason: Position matches expected template region.

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Corresponding Author [corresponding_author]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Corresponding Author' section must exist.
Found: Matched heading: "Corresponding author" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Corresponding Author' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, right-center area, upper area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: PASS

Graphical Abstract [graphical_abstract]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Graphical Abstract' component must exist.
Found: Matched heading: "Graphical abstract" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Graphical Abstract' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, upper area
Found: Page 1, left-center area, upper-middle area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Image should be present.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: PASS

English Abstract [abstract]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'English Abstract' section must exist.
Found: Matched heading: "Abstract" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'English Abstract' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, center area, upper-middle area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: PASS

Keywords [keywords]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Keywords' section must exist.

Found: Matched heading: "Keywords" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Keywords' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, lower-middle area
Found: Page 1, center area, upper-middle area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: PASS

Malay Abstract (Abstrak) [abstrak]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Malay Abstract (Abstrak)' section must exist.
Found: Nearby content: "78:1 (2016) 1–5 | www.jurnalteknologi.utm.my | eISSN 2180–3722 | Jurnal Teknologi Full Paper DESULFURIZATION OF HEAVY NAPHTHA
Confidence: 0.00
Detection Method: semantic_fallback
DOC Detection: NO
PDF Detection: YES
AI Detection: YES
AI Confidence: 0.70
AI Evidence: DOC extraction did not find the abstract section, showing only metadata and partial title information. PDF extraction found partial abstract content a
Evidence Sources: PDF + AI
Final Source Used: PDF + AI
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, lower area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Kata Kunci [kata_kunci]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Kata Kunci' section must exist.
Found: Detection method: not_found
Confidence: 0.00
Detection Method: not_found
DOC Detection: NO
PDF Detection: NO

AI Detection: NO

AI Evidence: DOC snippet contains title, authors, affiliations, article history, and abstract start but no keywords. PDF snippet similarly contains title, authors,

Evidence Sources: None

Final Source Used: None

Status: FAILED

Reason: Required component 'Kata Kunci' was not found.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, lower area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: FAILED

Introduction [introduction]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Introduction' section must exist.

Found: Matched heading: "1.0 INTRODUCTION" | Detection method: source_docx_text

Confidence: 0.85

Detection Method: source_docx_text

DOC Detection: YES

PDF Detection: NO

AI Detection: YES

AI Confidence: 0.85

AI Evidence: DOC snippet provides a detailed Introduction about sulfur compounds in petroleum fractions, environmental impact, and desulfurization methods. PDF sni

Evidence Sources: DOC + AI

Final Source Used: DOC + AI

Rejected false positives: 1.0 INTRODUCTION

Status: WARNING

Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Methodology [methodology]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Methodology' section must exist.

Found: Matched heading: "Materials and Methods" | Detection method: source_docx_text

Confidence: 0.85

Detection Method: source_docx_text

DOC Detection: YES

PDF Detection: YES

AI Detection: SKIPPED

Evidence Sources: DOC + PDF

Final Source Used: DOC + PDF

Status: PASS

Reason: The required component 'Methodology' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Page 1, left-center area, lower area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.20).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: PASS

Results and Discussion [results_discussion]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Results and Discussion' section must exist.
Found: Detection method: not_found
Confidence: 0.00
Detection Method: not_found
DOC Detection: NO
PDF Detection: NO
AI Detection: NO
AI Evidence: DOC snippet includes title, authors, affiliations, article history, and partial abstract only; PDF snippet includes similar metadata and partial graph
Evidence Sources: None
Final Source Used: None
Status: FAILED
Reason: Required component 'Results and Discussion' was not found.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: FAILED

Conclusion [conclusion]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Conclusion' section must exist.
Found: Detection method: not_found
Confidence: 0.00
Detection Method: not_found
DOC Detection: NO
PDF Detection: NO
AI Detection: NO
AI Evidence: DOC snippet contains title, authors, affiliations, and abstract start but no conclusion. PDF snippet contains title, authors, affiliations, article hi
Evidence Sources: None
Final Source Used: None
Status: FAILED
Reason: Required component 'Conclusion' was not found.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: FAILED

Acknowledgement [acknowledgement]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Acknowledgement' component is recommended.
Found: Detection method: not_found
Confidence: 0.00
Detection Method: not_found
DOC Detection: NO
PDF Detection: NO
AI Detection: SKIPPED
Evidence Sources: None
Final Source Used: None
Status: PASS
Reason: The required component 'Acknowledgement' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: PASS

References [references]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'References' section must exist.
Found: Detection method: not_found
Confidence: 0.00
Detection Method: not_found
DOC Detection: NO
PDF Detection: NO
AI Detection: SKIPPED
Evidence Sources: None
Final Source Used: None
Status: FAILED
Reason: Required component 'References' was not found.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: FAILED

Editor Conclusion

The manuscript does not fully comply with the template because one or more required items are missing or critically mismatched. The author should revise the manuscript using the official Jurnal Teknologi template.

Structure Score: 40% (Weight: 40%)

Layout Score: 89% (Weight: 30%)

Formatting Score: 100% (Weight: 30%)

Final Score: 73/100

Critical Items Failed:

• Results and Discussion — FAILED
• Conclusion — FAILED
• References — FAILED

Items Needing Review:

• Article Title — WARNING
• Authors — WARNING
• Affiliations — WARNING
• Malay Abstract (Abstrak) — WARNING
• Introduction — WARNING

This report was automatically generated by Turnitin Assistant v5.7.1 Template Screening Engine (DOCX-First + Multi-Source AI Evaluator).

Jurnal
Teknologi

Full Paper

DESULFURIZATION OF HEAVY NAPHTHA WITH SYNTHESIZED ACTIVE CARBON (PDAC) AND IMPREGNATED (CU_2CL): OPTIMIZATION, KINETIC, AND ISOTHERM STUDIES

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Fatima K.Hamza, Al-Diwaniyah, Iraq
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XB Month 202Y

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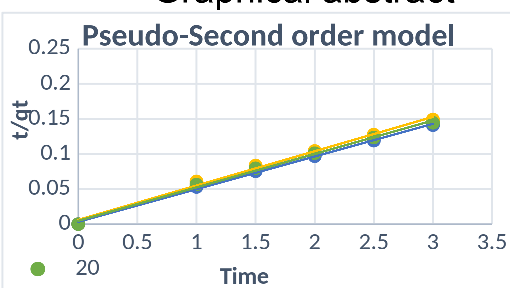
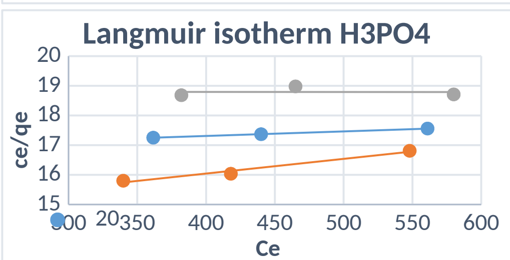
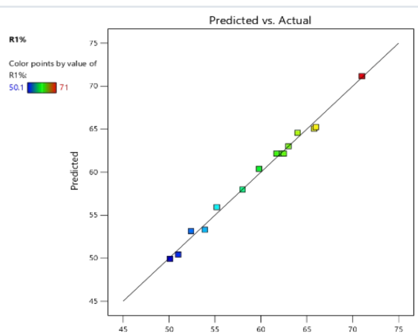
Please provide an **official organisation email** of the corresponding author

Graphical abstract







Abstract

This research explores the adsorptive removal of sulfur from heavy naphtha using prepared activated carbon from Iraqi date pits (Phoenix dactylifera L. cv. Zahidi) (PDAC) and copper chloride-impregnated date pits activated carbon (Cu_2Cl/PDAC). The adsorbent was optimized by Response Surface Methodology (RSM) using Box-Behnken Design to study the influence of temperature, contact time, and adsorbent dosage on the sulfur removal efficiency. The optimum conditions led to a maximum sulfur removal efficiency of 71% (31°C, 2.96 hr, and 0.0397 g/mL adsorbent). Kinetic studies revealed that sulfur adsorption followed the pseudo-second-order model, suggesting chemisorption as the adsorption mechanism, and equilibrium data were best fitted with Freundlich model of adsorption isotherms based on temperature. The research confirms that low-cost biomass-based activated carbon impregnated with Cu_2Cl is a viable and promising adsorbent to remove sulfur compounds from heavy naphtha, which helps in producing clean fuel and protects the environment.
Keywords: Heavy naphtha; Adsorptive desulfurization; PDAC; Response Surface Methodology; Kinetics; Isotherm models.

1.0 INTRODUCTION

Petroleum fractions, particularly heavy naphtha, contain sulfur compounds that are significant environmental pollutants as they are oxidized to sulfur oxides (SO_2) during combustion, leading to acid rain, air pollution and catalyst poisoning [1, 2]. The world's increasingly stringent environmental regulations have increased the demand for ultra-deep desulfurization processes to supply cleaner fuels[1,3]. While traditional hydrodesulphurization processes are commercially mature, they are typically associated with high costs, harsh operating conditions and low yields for recalcitrant sulfur compounds such as thiophenic compounds[2,3]. Adsorptive desulfurization has gained popularity as an alternative method owing to its low energy consumption, ease-of-operation and selective removal of sulfur-containing compounds at mild conditions[4]. Activated carbon is a promising adsorbent due to its high specific surface area, well-developed pore structure, chemical inertness, and low cost[4]. In recent years, bio-based activated carbons have attracted special attention as environmentally-friendly adsorbents due to their use of renewable agricultural waste materials and low cost[4]. Iraq and other Middle Eastern nations have a significant amount of date pits, an agricultural waste

Linear Equation:

$$\frac{t}{qt}=\frac{1}{K_2 q2e}+\frac{1}{qe}$$

Where: **k_2:** Pseudo-second order rate constant (g/mg·min).

In most liquid phase adsorption studies, this model has a very high correlation coefficient (R^2).

3.0 ISOTHERM MODELS

These models refer to the equilibrium relationship between the concentration of the substance in solution and its amount on the adsorbent at a constant temperature[13].

3.1 Langmuir Isotherm

Model This is based on the adsorption of a single layer of molecules on a uniform surface, with no interaction between adsorbed molecules [14].

Linear Equation:

$$\frac{C_e}{q_e}=\frac{1}{K_L q_m}+\frac{C_e}{q_m}$$

Where: **Ce:** Concentration of the substance at equilibrium (mg/L).

q_m: Maximum monolayer adsorption capacity (mg/g).

K_L: Langmuir constant (related to energy of adsorption).

3.2 Freundlich Isotherm

Model This empirical model is used to describe the adsorption on a heterogeneous surface and allows formation of multilayers [15].

Linear Equation:

$$\log q_e=\log K_F+\frac{1}{n}\log C_e$$

Where**: K_F:** Freundlich constant related to the capacity of adsorption.

1/n: Factor related to the intensity of adsorption and the heterogeneity of the surface

product, which is a cheap and sustainable raw material for the preparation of activated carbon due to its high carbon content. Activation with phosphoric acid can increase the carbon's porosity and surface functional groups, which boost the sulfur adsorption [4]. Moreover, the impregnation of various metals (e.g. copper chloride) can improve adsorption capacity by creating more adsorption sites and enhancing the interactions with sulfur compounds[5,6]. Design of experiments (DOE) using statistical methods such as Response Surface Methodology (RSM) and Box-Behnken Design are powerful methods to assess process variables and interactions with reduced experimental effort[7,8]. Recent local investigations on Iraqi refinery streams demonstrated that adsorptive desulfurization with locally sourced adsorbents is feasible, highlighting the need for sustainable fuel purification systems [9]. Hence, this study deals with the preparation, optimization, kinetic and thermodynamic studies of Cu_2Cl-loaded date pit activated carbon for the desulfurization of heavy naphtha, presenting an effective and sustainable method of preparing environmentally-friendly fuels locally.

2.0 KINETIC MODELS

Kinetic models are used to find the effectiveness of the adsorbent and the time needed to attain equilibrium. The most common models are:

2.1 Pseudo-First Order Model

This model was first introduced by Lagergren in 1898 and it suggests that the rate of adsorption is controlled by the availability of adsorption sites on the adsorbent's surface [10].

Linear Equation:

$$\ln(q_e - q_t)=\ln q_e - k_1 t$$

Where: **qe and qt**: Amount of adsorbate at equilibrium and at time t (mg/g).

k_1: Pseudo-first order rate constant (min^{-1}).

2.2 Pseudo-Second Order Model

This model assumes that the adsorption is chemisorption, meaning that there are valence forces through the sharing or exchange of electrons between the adsorbent and adsorbate [11,12]

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4.0 MOTHODLOGY

4.1 Materials and Methods

The date pits used in this research were sourced from the date palm (Phoenix dactylifera L. cv. Zahidi) located on local farms in Diwaniyah, Iraq. The samples were thoroughly washed with distilled water to remove any remaining impurities, then sun-dried for three days, followed by oven-baking at 110°C for three hours. The dried material was then ground and sieved to obtain particles ranging in size from 150 to 300 micrometers.

A type of adsorbent, acid-activated carbon (PDAC), was synthesized. For carbonization, 100 grams of the produced PD date pits were heated at 600°C for two hours under a continuous nitrogen flow rate of 3 liters per minute.

45 g of pre-carbonized material was impregnated with 300 mL of 4 M phosphoric acid solution (carbon-to-solution ratio 1:6.7 w/v). The mixture was stirred for 30 minutes and then left to infuse for 24 hours before drying. The impregnated samples were activated at 600 °C for 2 hours in a nitrogen atmosphere, then rinsed with deionized water to neutral pH and dried at 110 °C for 6 hours.

The activated carbon was impregnated with copper(II) chloride solution, prepared by dissolving 0.3 g of $CuCl_2$ in 40 mL of distilled water per 5 g of activated carbon. The impregnation ratio was 6 wt% ($CuCl_2$/carbon). The mixture was stirred at 60 °C for 5 hours to ensure uniform distribution of the metallic precursor across the entire carbon network. The suspension was then filtered, and the resulting solid precipitate was dried at 100 °C and calcined for 2 hours in a nitrogen atmosphere to prepare mineral-loaded activated carbon. Adsorption experiments were conducted using heavy naphtha produced at the Diwaniyah refinery, which had an initial sulfur content of 1200 ppm. The mineral-loaded activated carbon was the adsorbent used in the batch adsorption tests as show in figure (1). Quantities of adsorbent (2, 3, and 4 grams) were added to 100 mL of naphtha, and the mixtures were analyzed at different temperatures (20, 30, and 40 °C). The mixtures were mixed under controlled conditions, and liquid samples were withdrawn at specified time intervals. The amount of residual sulfur was determined to measure the desulfurization efficiency. To ensure the accuracy of elemental measurements, the sulfur content after adsorption was determined using X-ray fluorescence spectroscopy (XRF))NEXQC,Rigaku Corporation,Japan). The adsorption kinetics were analyzed using pseudo-first-order (PFO) and pseudo-second-order (PSO) kinetic models to explain the adsorption mechanism and identify the phases controlling its rate. The kinetic parameters were determined using the linear formulas of the different models and evaluated using correlation coefficients (R^2).

The experimental data were fitted to the Langmuir and Freundlich isotherm models to describe the equilibrium adsorption phenomenon. The parameters of the models were determined from linear plots and correlation coefficients (R^2).

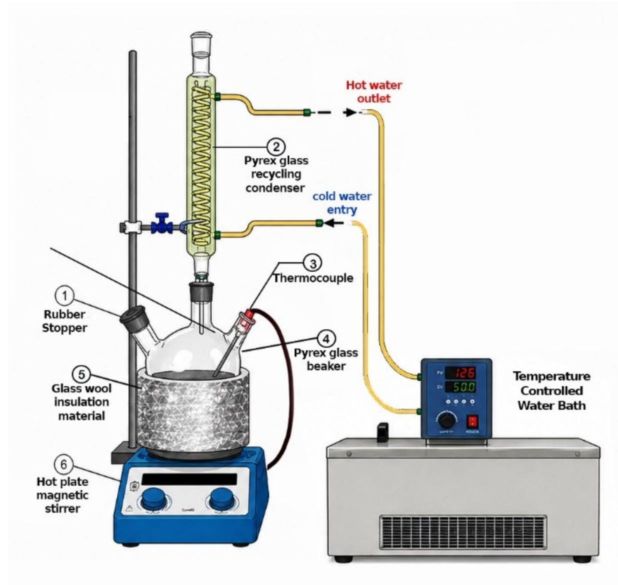


Figure (1): the adsorption process apparatus

4.2 Process parameter optimization

In order to investigate the influence of process variables and their interaction in the process of adsorptive desulfurization, a series of experiments was performed, varied the key operating variables; temperature, contact time and adsorbent dosage. The sulfur content of the feed was maintained at 1200 ppm using heavy naphtha from Al-Diwaniyah refinery. The Box-Behnken Design (BBD) experimental design involved 15 experiments with three independent variables (temperature (A), contact time (B), adsorbent loading (C) Three levels (low, medium and high) of the variables were investigated: 20-40 C, 1-3 hours and 0.02-0.04 g ml-1, for temperature, contact time and adsorbent dosage, respectively. The design in Table (1) is a combination of factorial and repeated center points (runs 2, 6 and 13) and is used to estimate experimental error and the reproducibility of the model. The central point (30 C, 2 h, 0.03 g mL -1) is the center point and represents the average of all variables and it is used to check the curvature of the response surface.

Std	***Run***	***Factor 1***	***Factor 2***	***Factor 3***
		A:Temp. C	**B:Time hour**	**C:Conc. gm/ml**
3	1	20	3	0.03
14	2	30	2	0.03
8	3	40	2	0.04
11	4	30	1	0.04
2	5	40	1	0.03
15	6	30	2	0.03
1	7	20	1	0.03
12	8	30	3	0.04
9	9	30	1	0.02
6	10	40	2	0.02
7	11	20	2	0.04
4	12	40	3	0.03

13	13	30	2	0.03
10	14	30	3	0.02
5	15	20	2	0.02

The varying of the experimental conditions in the experiments helps in assessing the main effects and interaction effects of the variables. Runs 10 and 15 assess the effect of temperature with constant contact time and dosage of the adsorbent, while run 3 and 11 assess the interaction effect of extreme temperatures and dosage with a constant contact time. This systematic design allows the development of a quadratic model for the adsorption-desulfurization process and in the determination of optimum conditions with lesser number of experiments than the full factor design.

6.0 RESULTS AND ANLAYSIS

6.1 RSM statistical analysis and results of the experiment

The RSM is a statistical method that was included in the experimental design to determine the effect of process parameters (independent variables) on a single response variable or multiple variables. (dependent variables). RSM effectively reduces the amount of experiments required to be conducted and it takes into account the interactions between the parameters [7].The desulfurization of heavy naphtha was performed in two stages by 15 batch tests. of different process parameters. These are presented in Table 2. The desulfurization efficiencies were determined with actual and predicted values. There was a good level of agreement between the actual and estimated results as shown in Figure 2. Desulfurization efficiencies were calculated by Design-Expert 13 software[16]. to determine the relationship between the percent desulfurization and the process parameters. The final quadratic equations which were the results of analysis are as follows:

$$\mathbf{R1\% = 62.17 + 0.5125A + 4.05B + 6.56C + 1.83AB - 0.85AC + 0.025BC - 2.83A^2 - 0.6583B^2 - 0.9833C^2} \quad \mathbf{(5)}$$

In which, R1% represents the desulfurization efficiency of heavy naphtha and (A, B, C) represent, respectively: A, B and C represent the temperature, time and amount of adsorbent used per 100 ml of heavy naphtha, respectively. The interaction effects of the model parameters are represented by AB, AC, and BC, and A2, B2 and C2 represent the quadratic constraints of the process parameters. The signs of the coefficients in Equation (5) suggest the direction of the effect of each process parameter. A positive coefficient represents a positive or additive effect, while a negative coefficient represents a negative or subtractive effect. The influence of each variable is determined by the overall effect of its linear, interaction and quadratic components[8].

Table (2): Experimental design values of response variables for the removal percentage of predicted and actual values

	Factor 1	***Factor 2***	***Factor 3***	***Removal %***	
Run	**A:Temp.**	**B:Time**	**C:Conc.**		
	C	**hour**	**gm/ml**	**Actual Value**	**Predicted Value**
1	20	3	0.03	59.8	60.39
2	30	2	0.03	62.5	62.17
3	40	2	0.04	64	64.58
4	30	1	0.04	63	63.01
5	40	1	0.03	53.9	53.31
6	30	2	0.03	61.7	62.17
7	20	1	0.03	55.2	55.94
8	30	3	0.04	71	71.16
9	30	1	0.02	50.1	49.94
10	40	2	0.02	52.4	53.15
11	20	2	0.04	66	65.25
12	40	3	0.03	65.8	65.06
13	30	2	0.03	62.3	62.17
14	30	3	0.02	58	57.99
15	20	2	0.02	51	50.43

6.2 Analysis of variance

Variance partitioning with sources of variance allowed an assessment of the suitability of the Box-Benkhen design [17]. Fisher's F-test was used to conduct an analysis of variance (ANOVA) to determine the significance of the coefficients[7]. The F-values and P-values for sulfur adsorption are shown in Table 3. The table shows the degrees of freedom, sum of squares, mean squares, F-value (Fisher's) and P-value. The P-test is carried out to check the F-value is high enough to affirm the statistical significance between the different statistical variables. The significance of the coefficient usually increases with decreasing P-value, but the significance of a factor increases with increasing F-value [8].The model boundaries were considered significant if P-values were less than 0.05 and non-significant if P-values were greater than 0.1[7].

Table 3 presents the ANOVA model for heavy naphtha desulfurization. The analysis of variance (ANOVA) model demonstrates a statistical significance with an F-value of **7**3.7**8** and a p-value below 0.0001. As seen in Table 4, The standard deviation is 0.8904. The R^2 value is 0.9925, the adjusted R^2 value is 0.9791, and the coefficient of variation (CV) is 1.49%, signifying an appropriate accuracy of 29.1946.

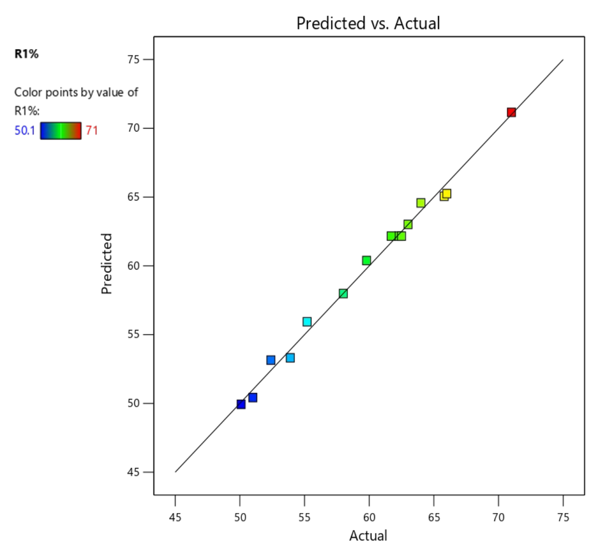


Figure (2): Plot of predicted vs. actual for % sulfur Removal

Table (3): ANOVA of response surface quadratic model for %sulfur Removal

Source	Sum of Squares	df	Mean Square	F-value	p-value
Model	526.44	9	58.49	73.78	< 0.0001
A-Temp.	2.1	1	2.1	2.65	0.1645
B-Time	131.22	1	131.22	165.51	< 0.0001
C-Conc.	344.53	1	344.53	434.56	< 0.0001
AB	13.32	1	13.32	16.8	0.0094
AC	2.89	1	2.89	3.65	0.1145
BC	0.0025	1	0.0025	0.0032	0.9574
A^2	29.64	1	29.64	37.39	0.0017
B^2	1.6	1	1.6	2.02	0.2146
C^2	3.57	1	3.57	4.5	0.0873
Residual	3.96	5	0.7928		
Lack of Fit	3.62	3	1.21	6.96	0.1283
Pure Error	0.3467	2	0.1733		
Cor Total	530.4	14			

Table (4): Summary of regression values for % sulfur Removal

Std. Dev	Mean	C.V. %	PRESS	R^2	Adjusted R^2	Predicted R^2	Adeq Precision
0.8904	59.78	1.49	58.66	0.9925	0.9791	0.8894	29.1946

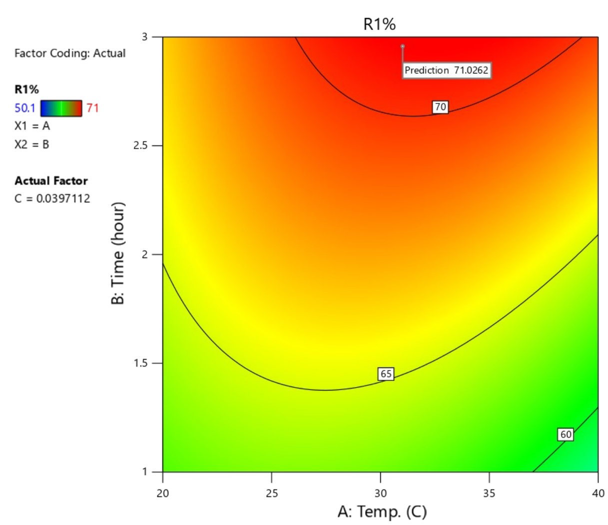
6.3 Response surface and contour plots for Desulfurization of Heavy Naphtha

the presented 3D response surface and 2D contour plots that demonstrate the influence of various parameters on the desulfurization efficiency (R1%) of heavy naphtha[9]. Such graphs are critical towards understanding conditions of ideal functioning[8].

In Figure (3), the desulfurization efficiency (R1%) of heavy naphtha was considerably improved by increasing the reaction temperature and time, resulting in a greater than 71% removal of sulfur. The response surface and contour plots demonstrate a synergistic effect between temperature and time, where maximum sulfur removal occurs at moderate temperatures and longer times. This trend can be explained by the enhanced mass transfer and kinetics at optimum operating conditions. This observation was also made by [2], who showed that temperature and contact time are important factors for enhancing deep desulfurization of fuel streams. Similarly, [3] report that improved reaction conditions also enhance sulfur removal from heavy streams. In addition, [1] highlighted that Response Surface Methodology (RSM) is a powerful statistical technique to determine and optimize desulfurization.

The interaction of temperature (A) and adsorbent concentration (C) on the sulfur removal (R1%) of heavy naphtha are shown in **Figure (4)**. The response surface and contour plots show that the optimal increase in both temperature and adsorbent concentration had a profound effect on sulfur removal efficiency, such that R1% improved from almost 50% to over 71%. Increasing adsorbent concentration increased the number of active sites, and increasing the temperature improved mass transfer and reaction rates, leading to better desulfurization. Nearly complete sulfur removal was achieved at medium-to-high temperature and maximum concentration[2 ,3, 1].

The response surface plot and contour plot in **Figure (5)** illustrates the interaction between reaction time (B) and adsorbent concentration (C) on the desulfurization efficiency (R1%) of heavy naphtha. It was found that higher reaction time and adsorbent concentration significantly enhanced sulfur removal, with R1% increasing from almost 50% at the low levels to more than 71% at the high levels. Increasing reaction time allowed more time for sulfur compounds to interact with the active sites of the adsorbent, and increasing adsorbent concentration provided a larger surface area for the sulfur species to adsorb. The high positive interaction between these factors suggests that increasing both of them is critical to maximize sulfur removal efficiency[2 ,3, 1].



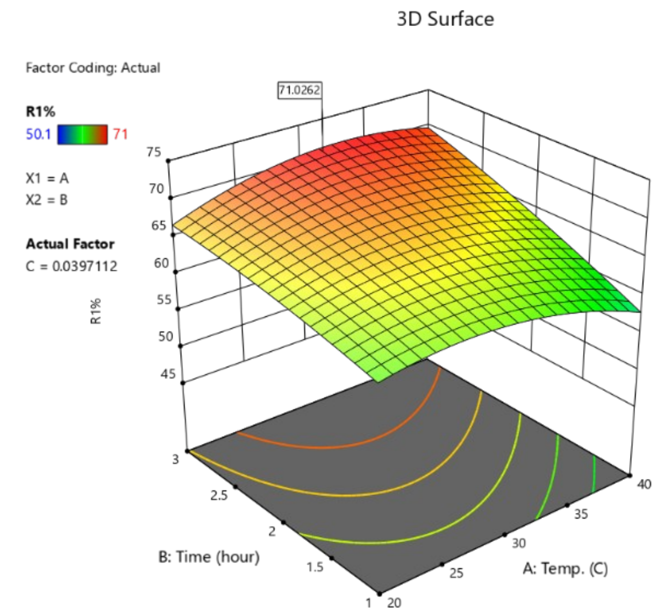
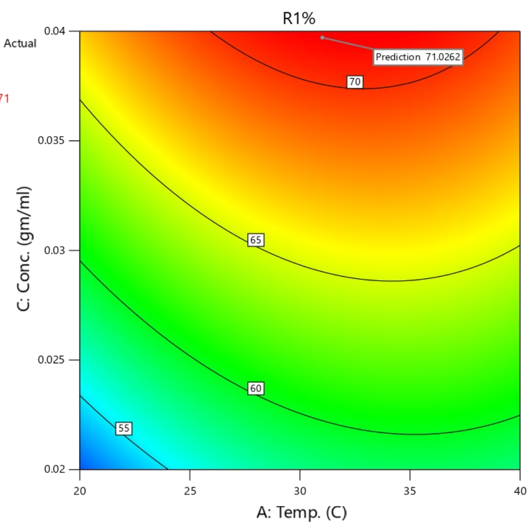


Figure (3) 3D and 2D plots of effect of Temperture and time on R1%



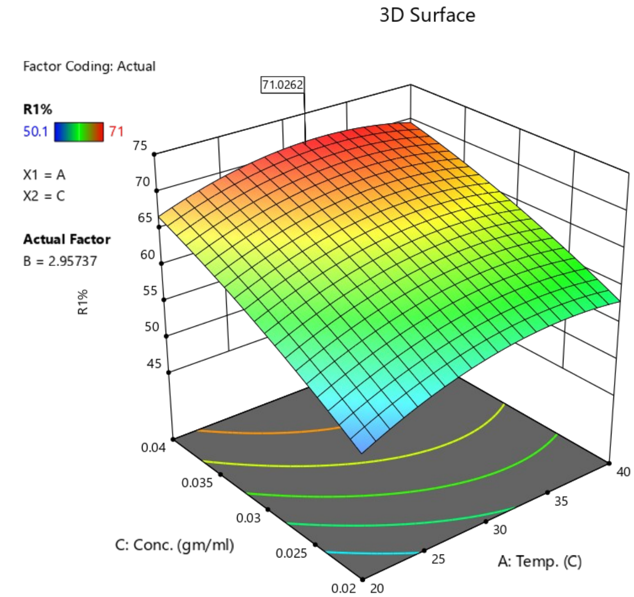
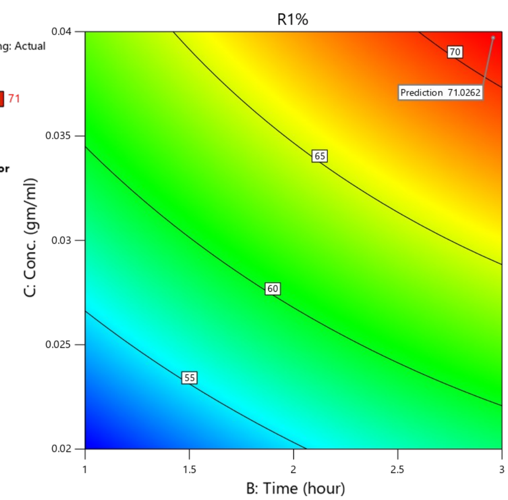


Figure (4): 3D and 2D plots of effect of Time and conc. on R1%



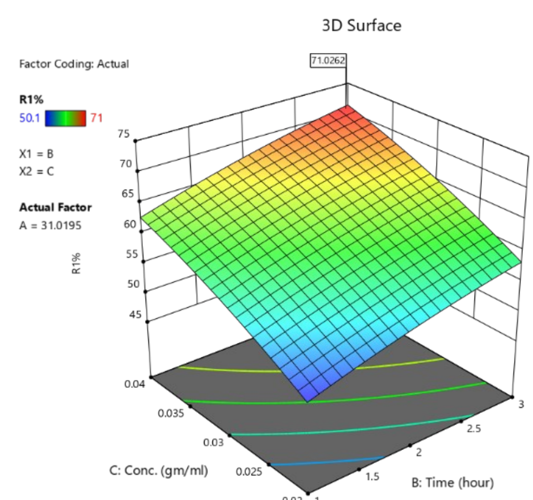


Figure (5) 3D and 2D plots of effect of Temperture and conc. on R1%

Table (5): Process parameters that gave the optimum removal efficiency

Parameters	*Temperture (C)*	*Time(hr)*	*Con.of AC (mg/ml)*	*% Sulfur Removal*
Optimum value	**31**	2.**9573**	0.0**3971**	**71**%

6.4 Fitting of Kinetic Models

In this study, the kinetics of the sulfur compounds from heavy naphtha onto the prepared adsorbent were determined under the best conditions. The adsorption kinetics were evaluated using the pseudo-first-order (PFO) and pseudo-second-order (PSO) kinetic models because of their simplicity and widespread use in understanding the adsorption process. The PFO model, suggested by Laguerre, is typically associated with physisorption, where the rate of adsorption depends on the availability of the adsorption sites[18]. However, the PSO model assumes that chemisorption is the rate-controlling step, which involves the sharing or exchange of electrons between adsorbent and sulfur compounds, thus providing a detailed description of the adsorption process[11].

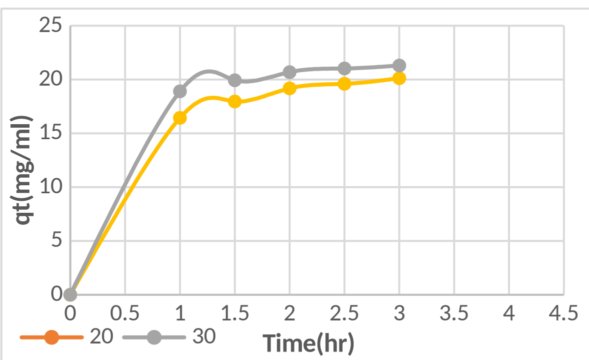


Figure (6): Rate of adsorption of sulfur on PDAC.

As shown in the figure (6), the adsorption capacity (qt) rapidly grew within the first 1.5 hours and then reached equilibrium. The equilibrium adsorption capacity dramatically reduced with the rise in temperature, in which the maximum sulfur uptake occurred at 30 °C._This suggests that the adsorption of sulfur compounds on the activated carbon is an exothermic reaction. Our results are in line with those reported on adsorptive desulfurization of petroleum fractions, which indicate that increasing the temperature results in a higher desorption rate, which in turn lowers the adsorption capacity [5,6].

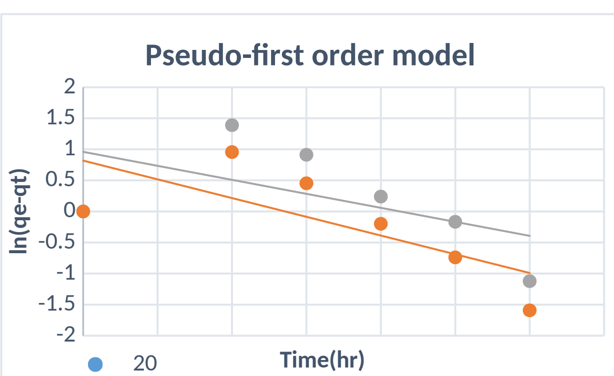


Figure (7). first order kinetic model for the adsorption of sulfur on PDAC at different Temperature

from the linear plots of ln(qe - qt) against time in figure (7)|. The results showed that the rate constant (k_1) was highest at 30 °C (0.6029 hr^{-1}), followed by 40 °C (0.4514 hr^{-1}) and 20 °C (0.1941 hr^{-1}). But the low values of the correlation coefficient (R^2 = 0.0917-0.5263) suggest that the pseudo-first-order model is not a good fit for the kinetics of the removal of sulfur. This implies that the adsorption process may be controlled by other mechanisms, such as chemisorption or intra-particle diffusion, which are better represented by other kinetic models [5,6].

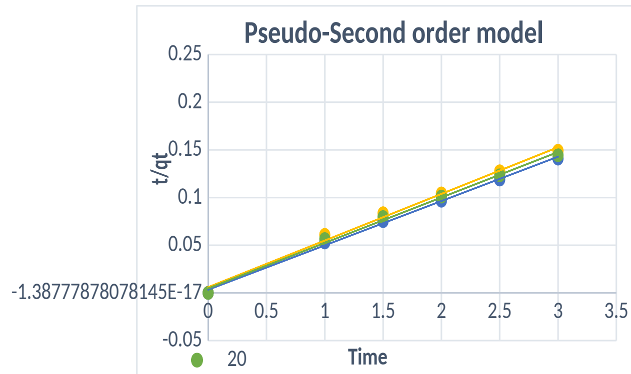


Figure (8). second order kinetic model for the adsorption of sulfur on PDAC at different Temperature

The pseudo-second-order model was also applied to the adsorption kinetics, with t/qt plotted against time. As displayed in the figure (8), this model showed a very good fit to the experimental data for all the temperatures with correlation coefficients (R^2) above 0.992. The R^2 value (0.9976) was the highest at 30°C as show in table (6) , indicating that the sulfur adsorption from heavy naphtha onto activated carbon is a pseudo-second-order process. This high correlation indicates that the rate-limiting step is chemisorption, which means the binding forces are valence forces through sharing or exchanging electrons between the adsorbent and the adsorbate. In addition, the calculated value of the equilibrium capacity (qe) using the reciprocal of the slope was found to be highest at 30°C, thereby confirming that 30°C is the most suitable temperature for desulfurization[5,6].

Table (6) : The values of R^2 and K for pseudo-second-order model

Temp. °C	***The pseudo-second-order model***	
	R^2	**K_2**
20	0.9953	0.4843
30	0.9976	0.6552
40	0.9927	0.3985

6.5 Fitting of Isotherm Models

The equilibrium isotherm data of the sulfur content in the naphtha and the adsorption on the activated carbon were described by the Langmuir and Freundlich models. The Langmuir model is based on monolayer adsorption on a surface with a finite number of homogeneous sites, whereas the Freundlich model is based on multi-layer adsorption on a heterogeneous surface. The separation factor (RL) and the Freundlich constant (1/n) were determined to assess the feasibility of the adsorption process, which suggests that the activated carbon has a good affinity for sulfur compounds under the experimental conditions [4,9].

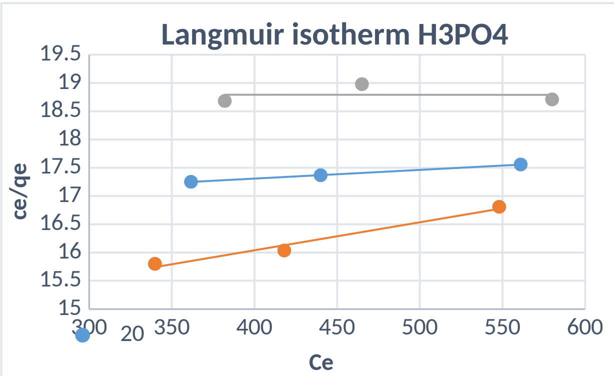


Figure (9). Langmuir adsorption isotherm model for sulfur adsorbed on the PDAC (at temperatures (20,30,40) °C.

The application of the Langmuir model to the desulfurization of heavy naphtha at various temperatures (20, 30 and 40 °C) is shown in Figure (9). The results demonstrated that at 20 °C, the model showed the highest correlation coefficient (R^2 = 0.9998), which means that the adsorption process is based on monolayer adsorption, where the surface is assumed to be energetically homogeneous. The maximum adsorption capacity (q) was calculated from the inverse slope and it was found that it decreased with increasing temperature and was highest at 20 °C. At 40 °C, the correlation coefficient (R^2 = 0.0001) was significantly lower, showing that the Langmuir model is inappropriate for describing the process at higher temperatures, and that the mechanism may change to a more complex system where surface homogeneity is not required[19].

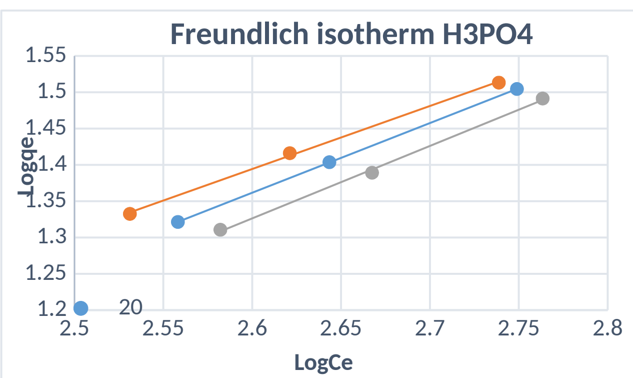


Figure (10). Freundlich adsorption isotherm model for sulfur adsorbed on the PDAC (at temperatures (20,30,40) °C.

The figure (10) describes the Freundich linear model of adsorption, where the results indicate a good agreement between the model and experimental data of all the temperatures studied with the correlation coefficient (R^2) close to one (0.9983 to 1.0). This indicates the "heterogeneity" of the surface of the activated carbon prepared with phosphoric acid and multilayer adsorption. The values of the adsorption intensity coefficient (n) have been calculated from the inverse slope (1/n) values and were found to be in the range (1.002-1.151) as shown in table (7) , which are all greater than one, suggesting that the adsorption of sulfur compounds on activated carbon is favourable under the experimental conditions. The proximity of the value of 1/n to one also suggests a uniform distribution of the energy of the active sites on the surface of the adsorbent[20].

Table (7) : The values of R^2 ,n and K_f for Freundlich adsorption isotherm model

Temp. °C	***Freundlich adsorption isotherm model***		
	R^2	**K_f**	**n**
20	0.9989	0.0732	1.0419
30	1	0.1369	1.1516
40	0.9983	0.05409	1.0021

7.0 CONCLUTION

Activated carbon (PDAC) impregnated with Cu_2Cl as an adsorbent for desulfurization of heavy naphtha and proved successful in the process. The response surface methodology (RSM) approach was used effectively to determine the optimal operating conditions, leading to 71% maximum sulfur removal efficiency under mild operating conditions. The kinetic study validated the pseudo-second-order model for sulfur adsorption, suggesting that chemisorption was the dominant adsorption mechanism. The isotherms indicated beneficial sulfur sorption with Freundlich model appropriately describing the multilayer sorption mode as a function of temperature.

This study underscores the potential of agricultural biomass waste, such as date pits, as a valuable source of adsorbents for petroleum refining, providing a cost-effective and eco-friendly alternative to traditional desulfurization processes. This process not only aids in the production of cleaner fuels but also facilitates waste utilization and resource sustainability in biomass-rich areas. Future research could explore adsorbent regeneration, scalability, and commercialization to enhance this technology.

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